

ARRAYS

Q) Alternating Rectangle Matrix Of Characters

DS	Session	Arrays	Question Information	Level 1 Challenge 22
Problem Description: Ram has provide inputs two numbers 'p' and 'q' to Sakthi. He wants to creates a matrix of size p x q (p rows and q columns) in which every elements is either Y or O. The Ys and Os must be filled alternatively, the matrix should have outermost rectangle of Ys, then a rectangle of Os, then a rectangle of Ys, and so on.. Constraints: 1 <= p, q <= 1000 Input Format: Input lines must be how many rows and columns in that matrix, also values must be separate space. Output Format: Print the output in a separate lines.				

CODE:

```
C main.c
1  #include <stdio.h>
2
3  int min(int a, int b) {
4      return (a < b) ? a : b;
5  }
6
7  int main(void) {
8      int p, q;
9      if (scanf("%d %d", &p, &q) != 2) {
10         return 0;
11     }
12
13     for (int i = 0; i < p; i++) {
14         for (int j = 0; j < q; j++) {
15             int layer = min(min(i, p - 1 - i), min(j, q - 1 - j));
16             if (layer % 2 == 0) {
17                 printf("Y ");
18             } else {
19                 printf("O ");
20             }
21         }
22         printf("\n");
23     }
24
25     return 0;
26 }
```

OUTPUT:

```
PS C:\Users\CH.SC.U4CSE25255\Projects\test> ./main
```

```
5 6
```

```
Y Y Y Y Y Y
```

```
Y 0 0 0 0 Y
```

```
Y 0 Y Y 0 Y
```

```
Y 0 0 0 0 Y
```

```
Y Y Y Y Y Y
```